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Developing a Culturally Adapted Adolescent Life-Skills Curriculum (THRIVE-Belize) Using Community-Based Participatory Research and AI-Assisted Synthesis

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In partnership with Toledo Community College (TCC) and Hillside Health Care, Toledo District, Belize

THE UNIVERSITY OF ARIZONA

— 01 THE NEED

Caribbean adolescents carry a heavy mental-health burden, and life-skills programs can help.

23.5%
of adolescents in the English-speaking Caribbean have a mental-health problem[1]

50
trials in 19 low- and middle-income countries show life-skills benefits[2]

In 2025, Hillside Health Care surveyed students at TCC. Risks clustered within the same students, and each of the seven planned modules matched a need the survey measured.

OUR QUESTION

Can AI agents help researchers find, read, and choose activities from long public manuals to build a curriculum for one school in the Toledo District?

— 02 WHO DID WHAT

MM
Mario Morales
- Designed the AI workflow and wrote its rules
- Set the inclusion criteria and, with Aimee, approved each module

AS
Aimee Slagle
- Checked random samples against the manuals
- Advised on cultural adaptation

GV
Gabriela Valdez
- Co-principal investigator who provided institutional support

AI agents
with three roles
- `Researcher` extracts every activity
- `Architect` checks it against the PDF
- `Local expert` flags culture, safety, cost

— 03 RULES THE AI COULD NOT OVERRIDE

- Right to pass, always
- Never promise secrecy
- Fictional scenarios only
- A quote and page for every activity
- Exactly 4 sessions per module

AI agents read 38 public manuals, more than 5,000 pages, so a small team could choose the best activities for one Belizean school.

— 04 FROM PAGES TO SESSIONS, JANUARY–FEBRUARY 2026

60 **records** from an AI web search of 27 queries, January 2026

5,074 **pages** in the 38 manuals the agents read in full

1,100+ **activities** catalogued, each with a quote and page

95 **activities** chosen for the THRIVE-Belize manual

How the agents worked, step by step

1 — Find (MM + AI)
In January 2026, the agent searched for free manuals for grades 6–12 with real classroom activities.

2 — Read (MM + AI)
Following Mario's rules, the agents cut each PDF into 10-page pieces and recorded every activity in 17 fields.

3 — Check (AS)
Aimee compared a random 20% of activities in the first manuals with the originals.

4 — Choose (AI + MM, AS)
The agents ranked activities by fit, and Mario and Aimee approved each module.

The result is 7 modules and 31 sessions of 40 minutes

M01	M02	M03	M04	M05	M06	M07
Communi-cation & emotions	Gender norms & boys' health	Sexual & reproductive health	Healthy relationships	Mental & physical health	Substance use & refusal	Environment & climate
S2-5	S6-9	S10-13	S14-17	S18-21	S22-25	S26-29

Session 1 opens the program, session 30 closes it, and a booster follows 8–12 weeks later. Every session is **paper-based and low-cost**.

— 05 ADAPTATION

We rewrote activities for Toledo and logged every change.

~~Generic food chain~~ → **Sardine, snapper, barracuda**
Fish that students know and eat. `Log #69`

~~Historical case study~~ → **Warning labels in six languages**
English, Kriol, Spanish, Q'eqchi', Mopan Maya, Garifuna. `Log #70`

~~Video scenarios~~ → **Read-aloud scenarios**
No projector is needed. `Log #30`

WHERE WE ARE NOW

We completed Phase 1 in July and August 2026. Students, teachers, parents, school leaders, and community members gave feedback in surveys, focus groups, and interviews, and we presented the manual to key stakeholders.

TAKE-HOME

With clear criteria and careful checks, AI agents let a small team read thousands of pages and build a curriculum in which every activity traces to its source.

WHAT WE LEARNED

Aimee's spot checks found activities the agents had skipped or partly invented. We then required them to extract every activity and to show that three counts matched.

A final read-through found timing errors that the automated checks had passed, and we logged them for correction.

Video and poster

AI disclosure. AI agents searched for manuals, extracted their activities, and drafted content under our written rules in January–February 2026. We checked their work against the original manuals and approved every module. AI also helped draft this poster, and we verified all of it.

Ethics. Phase 0 (2025 survey): Belize Ministry of Health and Wellness IRB GEN/147/01/25(42)Vol.VI. Phase 1 (2026): University of Arizona IRB STUDY00007955; Belize Ministry of Health and Wellness IRB MOHW/IRB/2026/002; Belize Ministry of Education authorization GEN/21/26(112) VOL. III; ClinicalTrials.gov NCT07687199.
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1. Liverpool S, Prescod J, Pereira B, Trotman C. Discov Ment Health 2023;3(1):11. doi:10.1007/s44192-023-00037-2 **2.** Singla DR, Waqas A, Hamdani SU, et al. Behav Res Ther 2020;130:103402. doi:10.1016/j.brat.2019.04.010

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